Systems and control: past paper practice

National 5 · Engineering contexts & challenges · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 Complete the diagram by adding the missing input energy, then state the type of control that uses feedback.

- (b) Kinetic energy (wind energy)
- (c) Closed loop control

2 Complete the sub-system diagram for the dishwasher with reference to the specification.

- External input labelled "set temperature"
- Heater (heating element) as the output block after the driver
- Temperature sensor in the feedback path
- Feedback arrows from the output to the sensor and from the sensor to the control

3 Complete the system diagram by adding the missing output, then state the type of control that has no feedback.

- (a) Sound
- (b) Open loop control

4a) Describe the reason for including a system boundary in a sub-system diagram.

- It shows which sub-systems are part of the system and which things are outside it (inputs and outputs).

4b) Describe, with reference to the sub-system diagram, the control of the water temperature.

- The water temperature is set by the user.
- The temperature sensor measures the actual water temperature and feeds it back to the control.
- The control compares the actual temperature with the set temperature.
- If it is too low, the output driver switches the heating element on; when it reaches the set temperature, it switches off.

5 Complete the system diagram by adding the missing output, then state the type of control that has no feedback.

- (a) Sound (noise)
- (b) Open loop control

6 Describe, with reference to the sub-system diagram, the operation of the system.

- The required water flow rate is set.
- The water flow rate sensor measures the actual flow rate and feeds it back to the control.
- The control compares the actual flow rate with the set flow rate.
- If they differ, the output driver switches on the motor, which drives the gear mechanism to open or close the gate.
- When the actual flow rate equals the set flow rate, the motor stops.

7 Complete the sub-system diagram for the closed loop motor speed control system.

- Speed sensor in the empty block
- Feedback arrows from the motor output to the speed sensor, and from the sensor back to the control unit

8 State the name of this type of diagram, and the function of the output driver.

- (a) Sub-system diagram (closed loop control diagram)
- (b) It amplifies (boosts) the signal from the control unit so that it can drive the heater.

9 Describe, using appropriate terminology, the operation of the wind turbine braking system.

- The maximum safe wind speed is set.
- The wind speed sensor measures the actual wind speed and feeds it back to the control.
- The control compares the actual wind speed with the maximum safe wind speed.
- If the wind speed is too high, the output driver operates the brake, producing a braking force that stops the turbine.